

Cristopher Miranda

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EDUCATION

Massachusetts Institute of Technology

Cambridge, MA | May 2027

Candidate for a Bachelor of Science in Mechanical Engineering | Zed Factor Fellow

GPA: 4.5/5.0

Selected Coursework: Thermal Fluids Engineering, Mechanics of Materials, Dynamics & Control, Electronics for Mechanical Systems, Design & Manufacturing, Measurement & Instrumentation

SKILLS

Hardware: Machining (Mill/Lathe, GD&T, CNC), Beckhoff PLC, Electrical Harnessing, Prototyping, Fixturing

Software: SOLIDWORKS/NX CAD, Ansys FEA, MATLAB, Python, Beckhoff TwinCAT, Structured Text (ST)

TECHNICAL EXPERIENCE

Vast

Long Beach, CA

Thermal Control Systems Test Engineering Intern

June 2026 – Present

- Designed and integrated fluid system test cart that uses compressed air to provide pneumatic back pressure into a thermal expansion tank to move up to 180 in³ of fluid in and out of flight accumulator for acceptance testing
- Created DAQ electrical box and HMI in TwinCAT using Beckhoff PLCs (4-20 mA AI, 2A DO, and TwinSAFE cards) and various electrical components to read and record sensor data, and safely control valves during testing
- Performed tests to measure leak rate under 10⁻⁶ scc/s GHe for pressurized K-port metal seal during vibration while joint is under preload, at bounds of nominal install torque, and during repeated reuse of seals

Varda Space Industries

El Segundo, CA

Automation & Test Engineering Intern

January 2026 – May 2026

- Designed and tested two-axis gimbaling mechanism (random positioning machine) using Beckhoff PLCs and two BLDC motors to create a simulated microgravity environment through the process of gravity vector averaging, resulting in reduced sedimentation for particle solutions, and a 14x reduction in effective gravity G over the course of one hour
- Built out electrical cabinet for random positioning machine containing 48V/24V PSUs, terminal blocks, 10A relays, slip rings, DC-DC converters, BLDC motors, and Beckhoff PLCs, with physical and electronic safety features to protect electronics and operators if motors stall and draw excess current
- Implemented random walk algorithm in TwinCAT XAE to Beckhoff EL7411 motor driver to randomize the direction of independently controlled BLDC motors, reducing effective gravity G by 50% compared to non-randomized motion

Starfish Space

Tukwila, WA

Thermal Systems Engineering Intern

June 2025 – August 2025

- Designed and integrated thermal vacuum chamber (TVAC) from end-to-end to perform in-house qualification and acceptance testing of various flight components for Starfish Space's Otter satellite servicing spacecraft
- Performed thermal analysis of TVAC system by quantifying expected thermal loads from test articles and ambient conditions to identify optimal cooling and heating solutions that minimize cost and are operable in high vacuum
- Implemented control algorithm and data logging features to follow user input temperature profiles from 30°C to +100°C with accuracy as high as ±0.1°C over 12 hours while uploading test data directly to the cloud

MIT Space Propulsion Laboratory

Cambridge, MA

Undergraduate Researcher - MIT Aeronautics & Astronautics

September 2025 – December 2025

- Manufactured PEEK emulator electrospray thruster bodies using precision machining techniques to create features $\leq$ 0.5 mm, and assembled related electronic components, to be integrated on a satellite slated for launch from the ISS in 2026
- Created fixtures, CNC programs, and manufacturing procedures to improve machining time and dimensional accuracy of emulator thruster bodies, ultimately reducing manufacturing time by 30% without sacrificing accuracy of parts

LEADERSHIP EXPERIENCE

MIT Society of Hispanic Professional Engineers

Cambridge, MA

Corporate Liaison - MIT Chapter

September 2023 – Present

- Managed corporate relationships between MIT-SHPE and industry partners, leading to 15 new corporate partnerships for 2025-2026, raising raising $\geq$ 35,000 and collaborating on various professional events with > 50 members

MIT Students for the Exploration and Development of Space

Cambridge, MA

Vice President - MIT Chapter

September 2024 – Present

- Coordinated logistics for events related to space and astronomy with the end goal of providing a non-technical area for MIT students to channel their interests in these fields through outings, guest lectures, or observational activities